

BARNO & ASHIQ'S PART

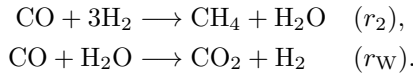
Governing balances, axial profiles, and operating-condition sensitivity for CO methanation.

Operating inputs, givens, and assumptions

Illustrative inlet (Figs. 1–2)	Dry feed $\text{CO}:\text{H}_2:\text{N}_2 = 1:4:25$ at 0.050 mol s^{-1} : $F_{\text{CO},0} = 0.001667$, $F_{\text{H}_2,0} = 0.006667$, and $F_{\text{N}_2,0} = 0.041667 \text{ mol s}^{-1}$; inlet CH_4 , H_2O , and CO_2 are zero. $T_0 = T_w = 623.15 \text{ K}$ (350°C), $P_0 = 5.00 \text{ bar}$.
Assumed bed and transport	$D_t = 0.0254 \text{ m}$, $L = 0.3048 \text{ m}$, and $\rho_B = 750 \text{ kg m}^{-3}_{\text{bed}}$ give $W_{\text{tot}} = 0.1158 \text{ kg}$ (calculated); $\varepsilon_b = 0.40$, $D_p = 3 \text{ mm}$, $\mu = 2.5 \times 10^{-5} \text{ Pa s}$, and $U = 10 \text{ W m}^{-2} \text{ K}^{-1}$. Pressure drop uses Ergun; wall heat exchange is enabled and fresh-catalyst activity is $a = 1$.
Model and numerical assumptions	Steady 1-D plug flow with the reduced M4 CO-methanation/WGS network; direct CO_2 methanation is omitted. Heat capacities are constant ($\text{CO}/\text{H}_2/\text{CH}_4/\text{H}_2\text{O}/\text{CO}_2/\text{N}_2$: $30.3/29.3/52.0/36.5/46.5/30.1 \text{ J mol}^{-1} \text{ K}^{-1}$) and reaction heats are fixed at $-\Delta H_2 = 206$ and $-\Delta H_W = 41 \text{ kJ mol}^{-1}$. The stiff system uses Radau ($\text{rtol} = 10^{-6}$) and 401 output points.
Group-14 reported inputs	Appendix B, Table 3: $\text{CO}/\text{H}_2/\text{N}_2 = 0.320/0.399/0.799 \text{ mol h}^{-1}$; catalyst = 3.00 g , reaction temperature = 350°C , and pressure = 760 mmHg (about 1.013 bar). These source-reported conditions are separate from the 5-bar illustrative run above.
Other figure cases	Figs. 3–4 vary inlet temperature ($250\text{--}400^\circ\text{C}$) and pressure ($1\text{--}15 \text{ bar}$, with Fig. 4 restricted to $5\text{--}15 \text{ bar}$), holding other base inputs fixed. Fig. 5 uses the Experiment 1 basis: 3.12 g catalyst, 350°C , a 1 bar pressure override, and isothermal operation; inlet H_2/CO ratio and catalyst space time vary. Figs. 6–7 use absolute feed-flow sweeps at 1 bar and 3.12 g catalyst. Figs. 8–9 present an isolated N_2 sweep and the four-factor grid-search optimization, respectively.

Model basis

The bed is treated as a steady, one-dimensional plug-flow reactor. The reduced network retains CO methanation and water-gas shift (WGS):



At each position, the reduced M4 kinetics give net rates r_2 and r_W from local temperature and partial pressures $p_i = y_i P$. Rates are in $\text{mol kg}_{\text{cat}}^{-1} \text{ s}^{-1}$, flows F_i in mol s^{-1} , and a is catalyst activity. Integration uses catalyst mass W . For uniform loading ρ_B in a tube of cross-section $A_c = \pi D_t^2/4$,

$$W = \rho_B A_c z, \quad z = \frac{W}{\rho_B A_c}, \quad W_{\text{tot}} = \rho_B A_c L.$$

A steady mole balance over a catalyst slice dW gives $dF_i/dW = a \sum_j \nu_{ij} r_j$, where a is catalyst activity. Written for the modeled species:

$$\begin{aligned}\frac{dF_{\text{CO}}}{dW} &= -a(r_2 + r_W), & \frac{dF_{\text{H}_2}}{dW} &= a(-3r_2 + r_W), \\ \frac{dF_{\text{CH}_4}}{dW} &= ar_2, & \frac{dF_{\text{H}_2\text{O}}}{dW} &= a(r_2 - r_W), \\ \frac{dF_{\text{CO}_2}}{dW} &= ar_W, & \frac{dF_{\text{N}_2}}{dW} &= 0.\end{aligned}$$

Temperature from the energy balance

For a differential slice, reaction heat raises the gas sensible enthalpy; heat transfer to the wall removes heat when the gas is hotter than the wall. Thus

$$\sum_i F_i C_{p,i} \frac{dT}{dW} = a[r_2(-\Delta H_2) + r_W(-\Delta H_W)] - \frac{U a_w}{\rho_B} (T - T_w).$$

Dividing by the gas heat-capacity flow gives the temperature equation integrated by the solver:

$$\frac{dT}{dW} = \frac{a[r_2(-\Delta H_2) + r_W(-\Delta H_W)] - \frac{U a_w}{\rho_B}(T - T_w)}{\sum_i F_i C_{p,i}}$$

Here $a_w = 4/D_t$ is the cylindrical heat-transfer area per bed volume. The formulation uses exothermic heats $-\Delta H_2 = 206$ and $-\Delta H_W = 41$ kJ mol⁻¹. The current implementation uses constant heat capacities and these constant reaction enthalpies.

Pressure from the Ergun equation

The Ergun pressure gradient combines viscous resistance (first term) and inertial resistance (second term):

$$\frac{dP}{dz} = - \left[\frac{150\mu(1 - \varepsilon_b)^2 u_s}{D_p^2 \varepsilon_b^3} + \frac{1.75\rho_g(1 - \varepsilon_b)u_s^2}{D_p \varepsilon_b^3} \right].$$

The reacting-gas properties are recalculated locally:

$$u_s = \frac{F_{\text{tot}} RT}{P A_c}, \quad \rho_g = \frac{P \bar{M}}{RT}, \quad \bar{M} = \sum_i y_i M_i, \quad y_i = \frac{F_i}{F_{\text{tot}}}.$$

Since $dz/dW = 1/(\rho_B A_c)$, the pressure equation in the solver's catalyst-mass coordinate is

$$\frac{dP}{dW} = - \frac{1}{\rho_B A_c} \left[\frac{150\mu(1 - \varepsilon_b)^2 u_s}{D_p^2 \varepsilon_b^3} + \frac{1.75\rho_g(1 - \varepsilon_b)u_s^2}{D_p \varepsilon_b^3} \right]$$

Pressure is integrated in Pa; the kinetic partial pressures are evaluated in bar. The pressure loss is coupled to reaction because the local flow, composition, temperature, and density change down the bed.

Integrated solution and plotting

The numerical problem is the coupled initial-value system

$$\frac{d\mathbf{y}}{dW} = \mathbf{f}(\mathbf{y}), \quad \mathbf{y}(W) = \begin{bmatrix} \mathbf{F}(W)^\top & T(W) & P(W) \end{bmatrix}^\top, \quad \mathbf{y}(0) = \begin{bmatrix} \mathbf{F}_0^\top & T_0 & P_0 \end{bmatrix}^\top, \quad 0 \leq W \leq W_{\text{tot}}.$$

The code integrates this system with the stiff Radau method and samples the solution at 401 evenly spaced catalyst-mass locations. The plotted coordinates are $z_j = W_j/(\rho_B A_c)$, $T_j = T(W_j)$, and $P_{j,\text{bar}} = P(W_j)/10^5$. CO conversion is calculated as

$$X_{\text{CO}}(W) = 1 - \frac{F_{\text{CO}}(W)}{F_{\text{CO},0}}.$$

Differentiating this definition and substituting the CO balance gives

$$\frac{dX_{\text{CO}}}{dW} = \frac{a(r_2 + r_W)}{F_{\text{CO},0}}.$$

Therefore, the figures show samples of the integrated profiles, not a separate closed-form solution for $T(z)$ or $P(z)$.

Plotted demonstration case. $D_t = 0.0254$ m (1.00 in), $L = 0.3048$ m (12.00 in), $\rho_B = 750$ kg m⁻³, $\varepsilon_b = 0.40$, $D_p = 3$ mm, $\mu = 2.5 \times 10^{-5}$ Pa s, $U = 10$ W m⁻² K⁻¹; $T_0 = T_w = 623.15$ K, $P_0 = 5$ bar, and feed ratio CO:H₂:N₂=1:4:25 at 0.05 mol s⁻¹. Several geometry, feed, transport, and heat-transfer inputs remain assumed; this run is illustrative, not experimental validation.

Equation basis: Subah and Somo formulation, Sections 3, 6, and 7. Numerical implementation: `src/methanation/reactor.py`; plotting: `src/methanation/reporting.py`.

CO conversion along the catalyst bed

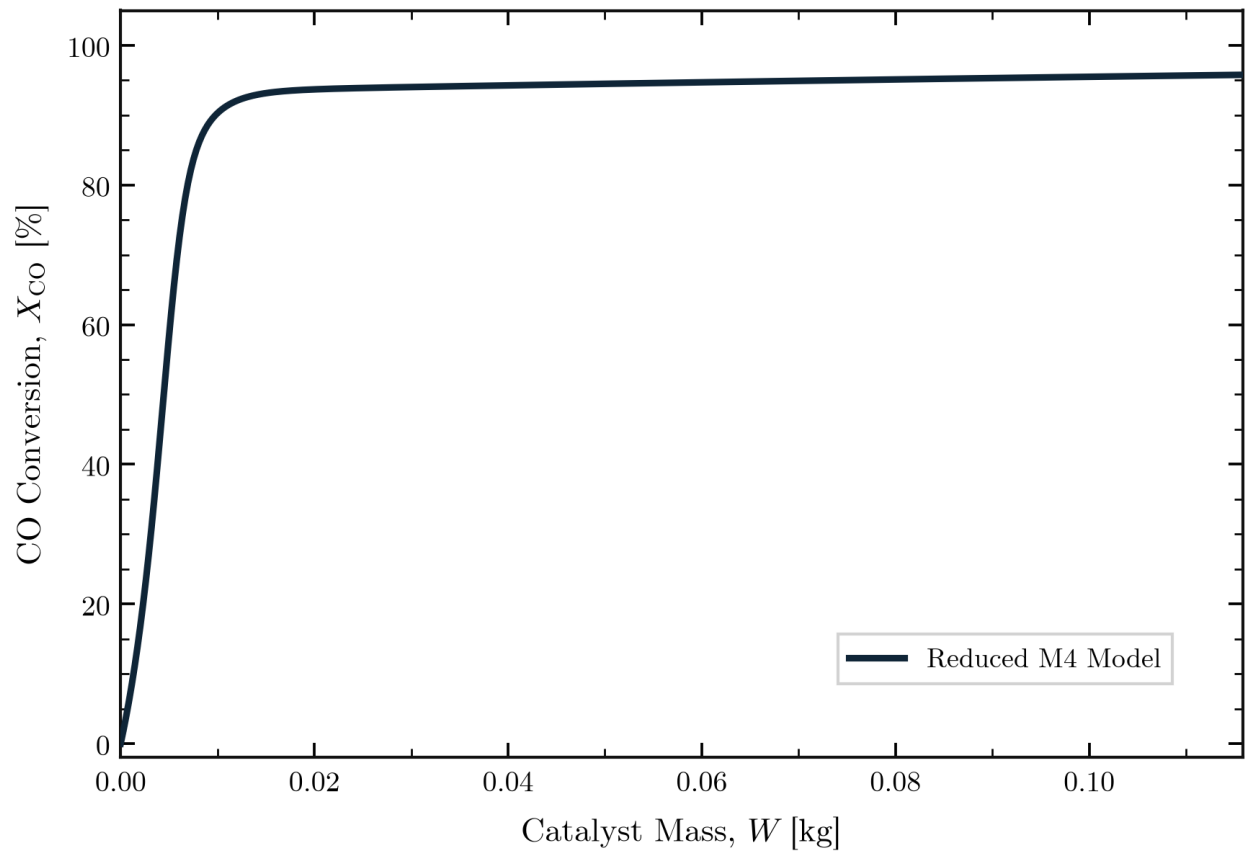


Figure 1. Reduced M4 model prediction of CO conversion versus catalyst mass for the demonstration case.

The calculated outlet conversion is 95.78% for this assumed configuration. This curve is a model demonstration, not an experimental fit or validation result.

Temperature and pressure along the bed

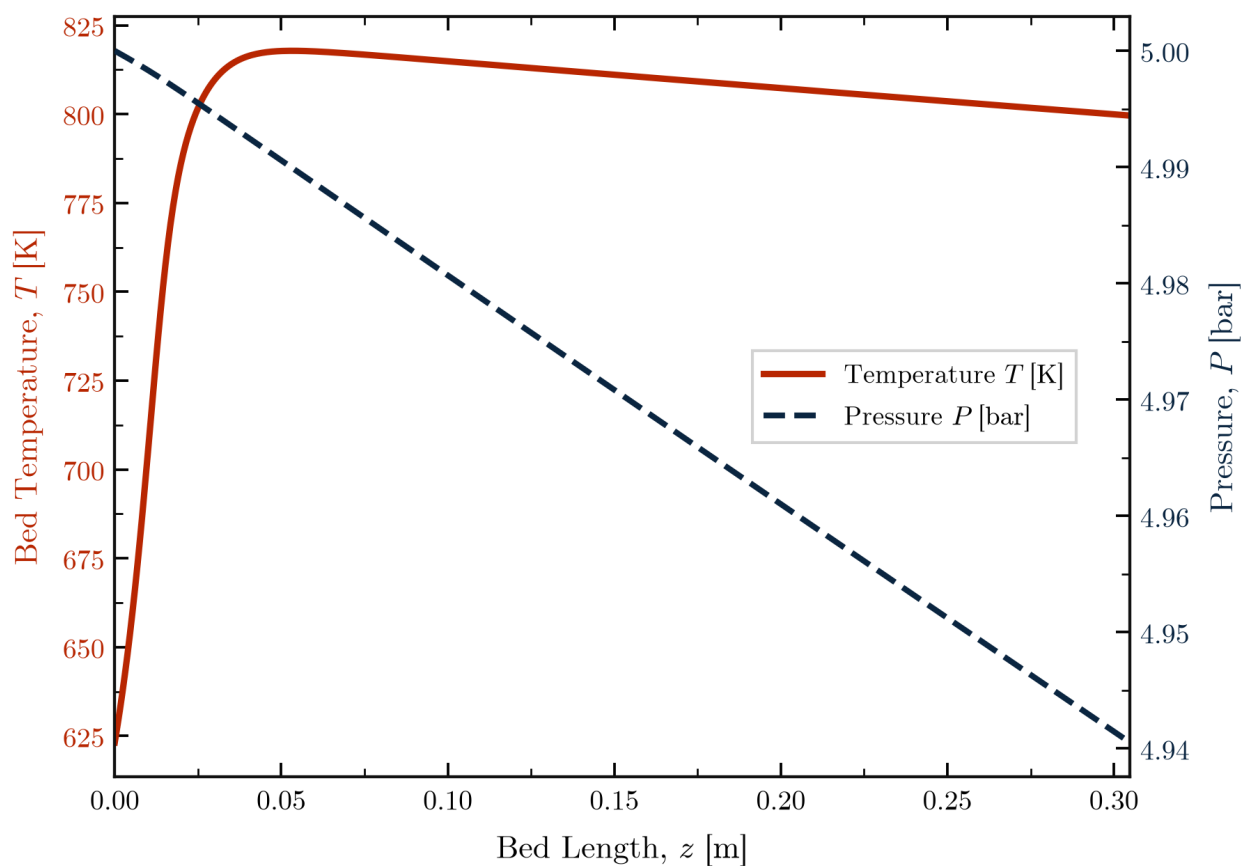


Figure 2. Integrated temperature and Ergun pressure profiles versus bed length.

The sampled temperature peaks at 817.8 K near $z = 0.0533$ m and is 799.6 K at the outlet. Pressure falls from 5.0000 to 4.9405 bar (about 5.95 kPa). The plotted pressure axis is tightly auto-scaled, so the visible line exaggerates this relative drop.

CO conversion across reactor inlet conditions

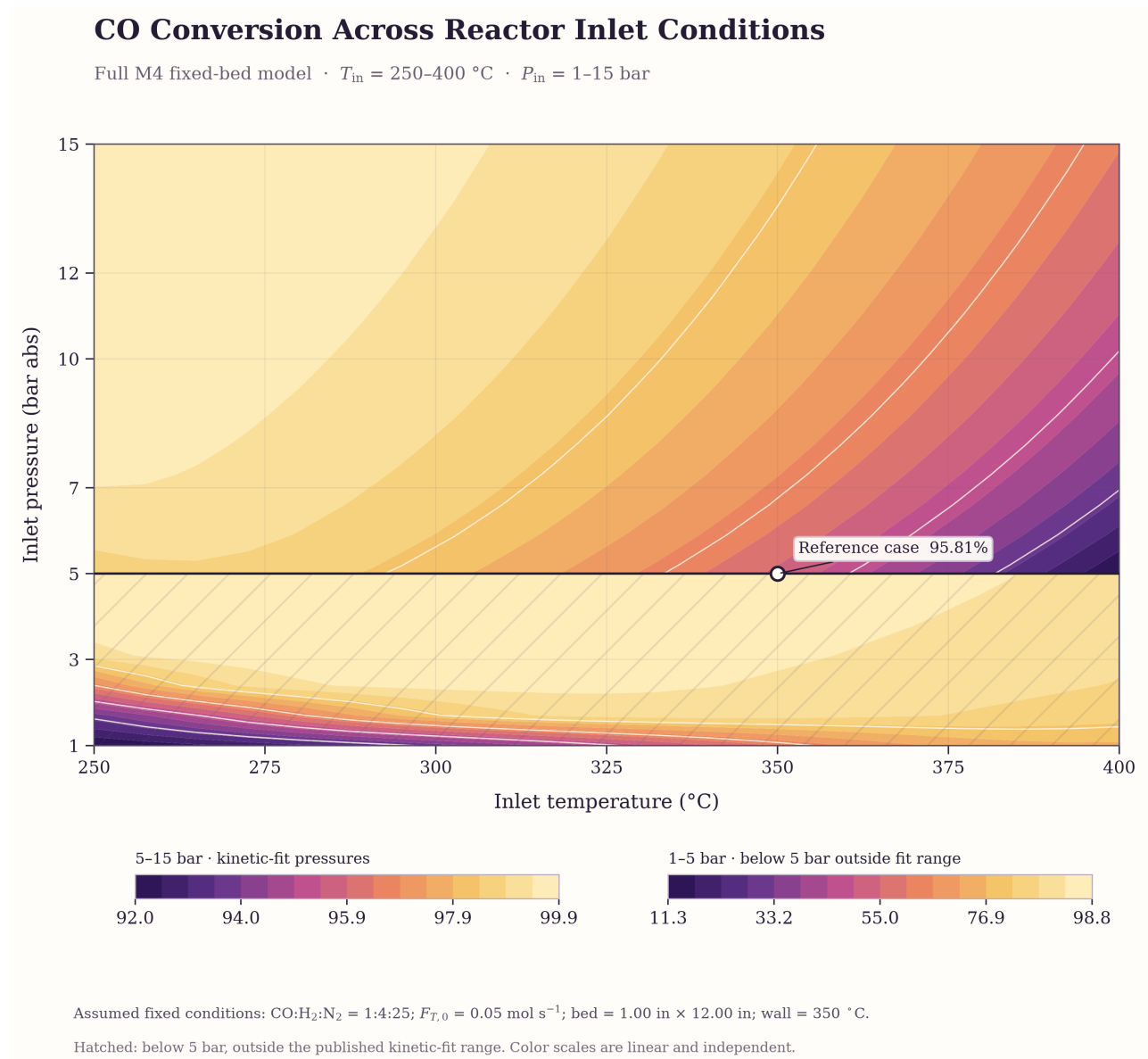


Figure 3. Full M4 sensitivity of bed-outlet CO conversion to inlet temperature and pressure.

The hatched $1 < 5\text{ bar}$ region is below the pressure range tested for the Celoria et al. kinetic fit and is an extrapolation; $5\text{--}15\text{ bar}$ is within the reported pressure range. The two bands use separate linear color scales, so read each colorbar independently. The sweep covers $250\text{--}400\text{ }^{\circ}\text{C}$ and varies only inlet temperature and pressure, with the other assumed base-case inputs fixed. Although inlet temperatures lie within the paper's fitted temperature range, 469 of 484 cases have sampled bed temperatures above $400\text{ }^{\circ}\text{C}$ and therefore extrapolate the kinetics locally. This is a conditional model sensitivity, not experimental validation. Kinetic source: [Celoria et al.](#)

CO conversion within the kinetic-fit pressure range

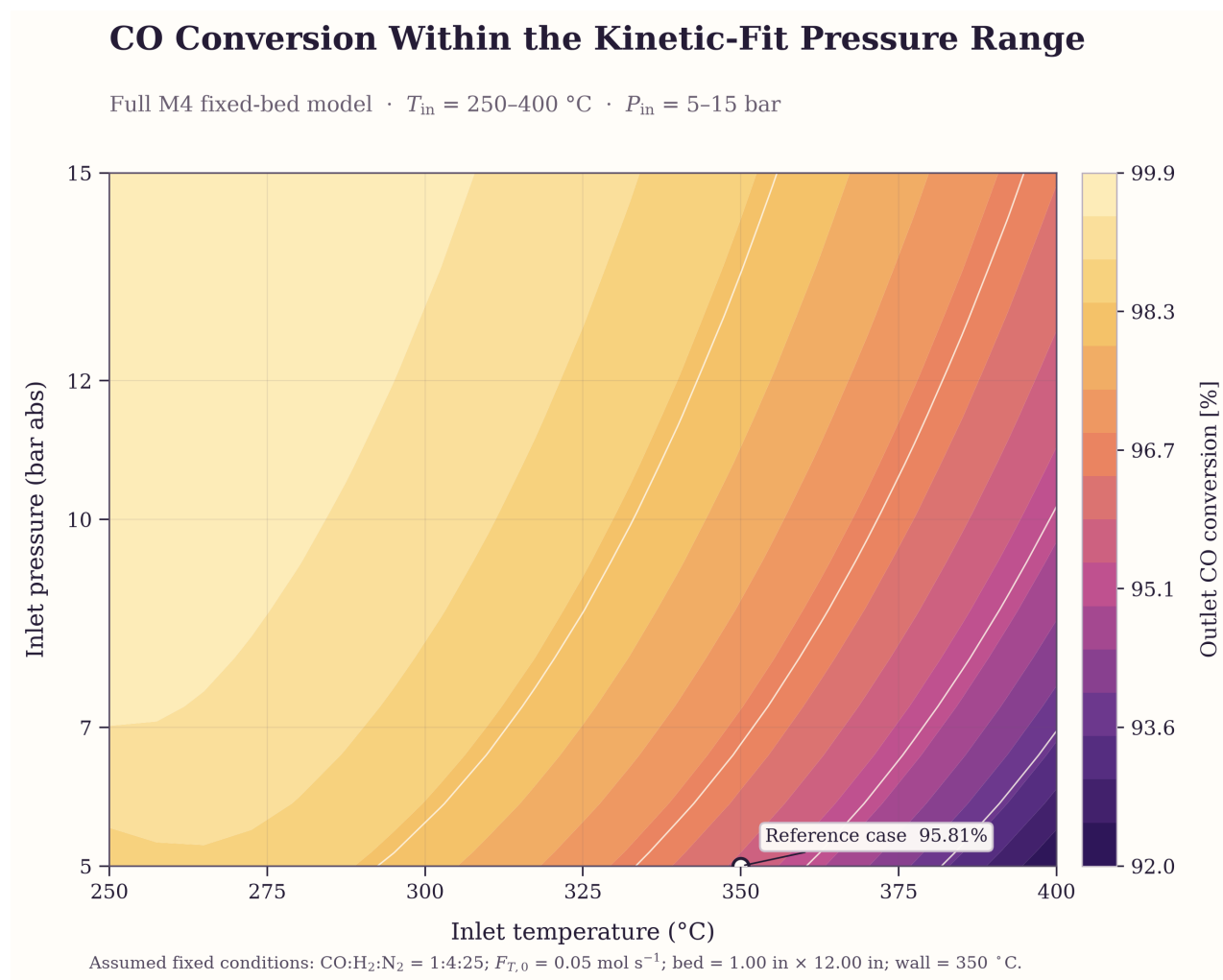


Figure 4. Full M4 sensitivity restricted to inlet pressures of 5–15 bar.

This view isolates the pressure interval reported for the Celoria et al. kinetic fit, with a single linear color scale. Inlet temperature spans 250–400 °C. The kinetic fit was tested at discrete temperature and pressure conditions; values between those conditions are model interpolations. The reactor's predicted local bed temperature may exceed 400 °C even when its inlet temperature is within range, so the result remains an exploratory conditional prediction rather than experimental validation.

CO conversion across inlet ratio and catalyst space time

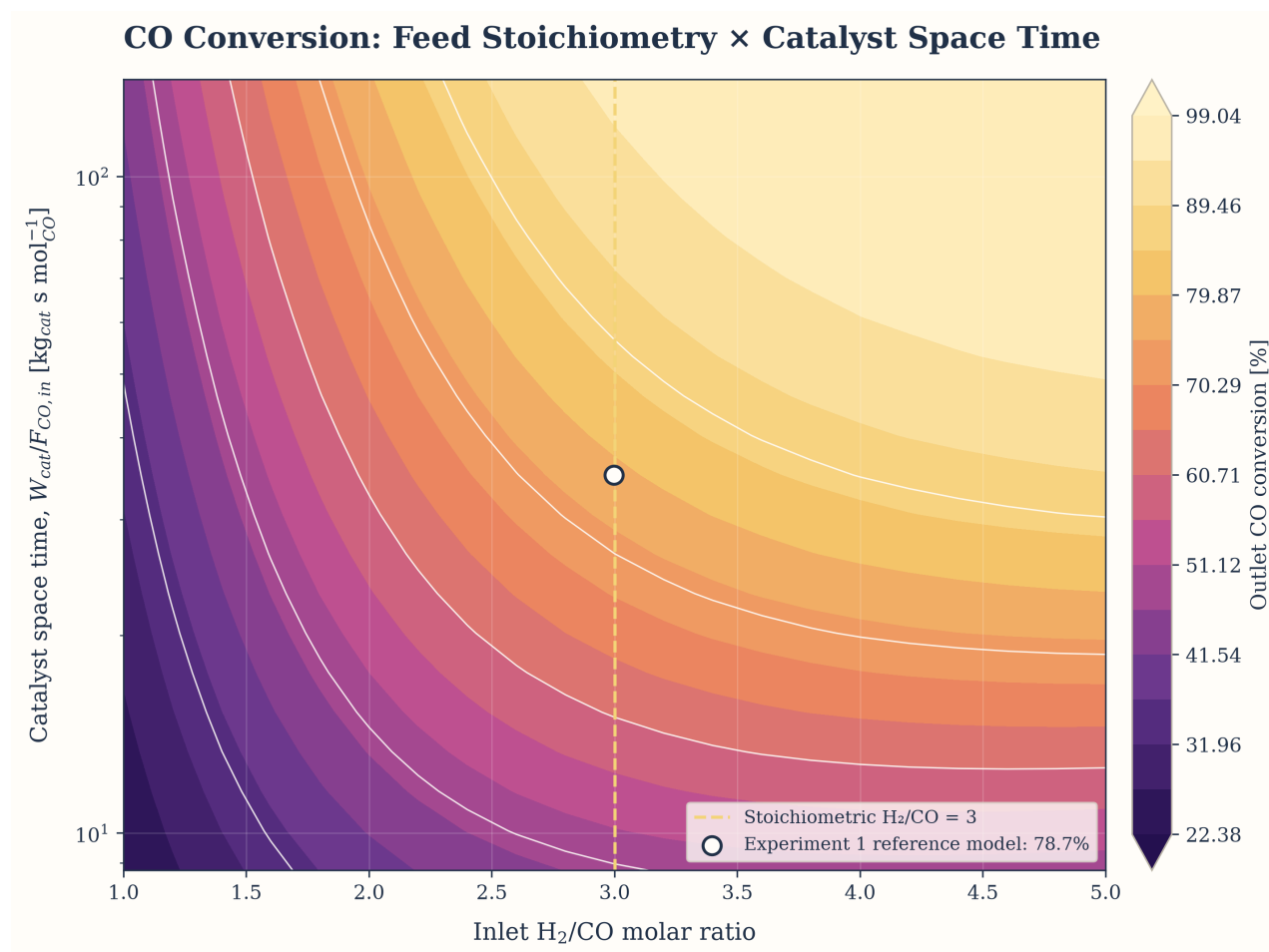


Figure 5. Full M4 predicted outlet CO conversion at 1 bar as inlet H₂/CO ratio and catalyst space time vary.

The circle identifies the modeled Experiment 1 reference point ($H_2/CO = 2.9969$, $W_{cat}/F_{CO,in} = 35.10 \text{ kg}_{cat} \text{ s mol}_{CO}^{-1}$), where predicted conversion is 78.72%. This differs from the source-reported 40.63%; the reported value is not used as a fit target. The sweep uses 1 bar absolute, consistent with the source's atmospheric-pressure description; Appendix A instead lists 2 bar. The selected pressure is below the 5–15 bar kinetic-fit range.

Defining the ratio and space-time sweep

The horizontal coordinate is the inlet molar ratio

$$r_{H_2/CO} = \frac{F_{H_2,in}}{F_{CO,in}},$$

where $r_{H_2/CO} = 3$ is the stoichiometric feed ratio for CO methanation. The vertical coordinate is catalyst space time on a CO-feed basis:

$$\boxed{\tau_{cat,CO} = \frac{W_{cat}}{F_{CO,in}}} \quad [\tau_{cat,CO}] = \text{kg}_{cat} \text{ s mol}_{CO}^{-1}.$$

Here W_{cat} is catalyst mass and $F_{CO,in}$ is inlet CO molar flow. This ratio has units of catalyst mass per CO throughput. A larger value means more catalyst is available per mole of CO fed. It is often informally called contact time, but it is not the gas residence time in seconds: gas residence time depends on bed void volume and the local volumetric flow, while $W_{cat}/F_{CO,in}$ is a catalyst-normalized throughput measure.

The sweep holds $W_{cat} = 3.12 \text{ g}$ fixed, so it changes space time by changing feed throughput. Equivalently,

$$F_{CO,in} = \frac{W_{cat}}{\tau_{cat,CO}}, \quad F_{H_2,in} = r_{H_2/CO} F_{CO,in}, \quad F_{N_2,in} = \left(\frac{F_{N_2}}{F_{CO}} \right)_{Exp1} F_{CO,in}.$$

Thus CO and N₂ flows scale together and H₂ is reset to the selected ratio; the N₂/CO ratio remains at its Experiment 1 value. All other selected inputs are held fixed at 350 °C, 1 bar absolute, and isothermal operation. The conversion plotted at each grid point is

$$X_{CO,out}(\%) = 100 \left(1 - \frac{F_{CO,out}}{F_{CO,in}} \right).$$

The map shows the combined conditional response to stoichiometry and catalyst per CO throughput. Conversion rises strongly as the feed approaches the stoichiometric ratio from below, then the improvement diminishes at higher H₂/CO and larger space time. These are model sensitivities, not validated operating predictions: the 1 in × 12 in geometry is assumed, the 3.12 g charge implies a sparse-bed voidage of about 0.98384 under the assumed particle density, and 1 bar is below the kinetic fit's 5 bar lower limit. The source describes atmospheric pressure in its abstract and discussion while Appendix A states 2 bar; this sweep follows the atmospheric-pressure description using a 1 bar override.

CO conversion across absolute reactive-feed flows

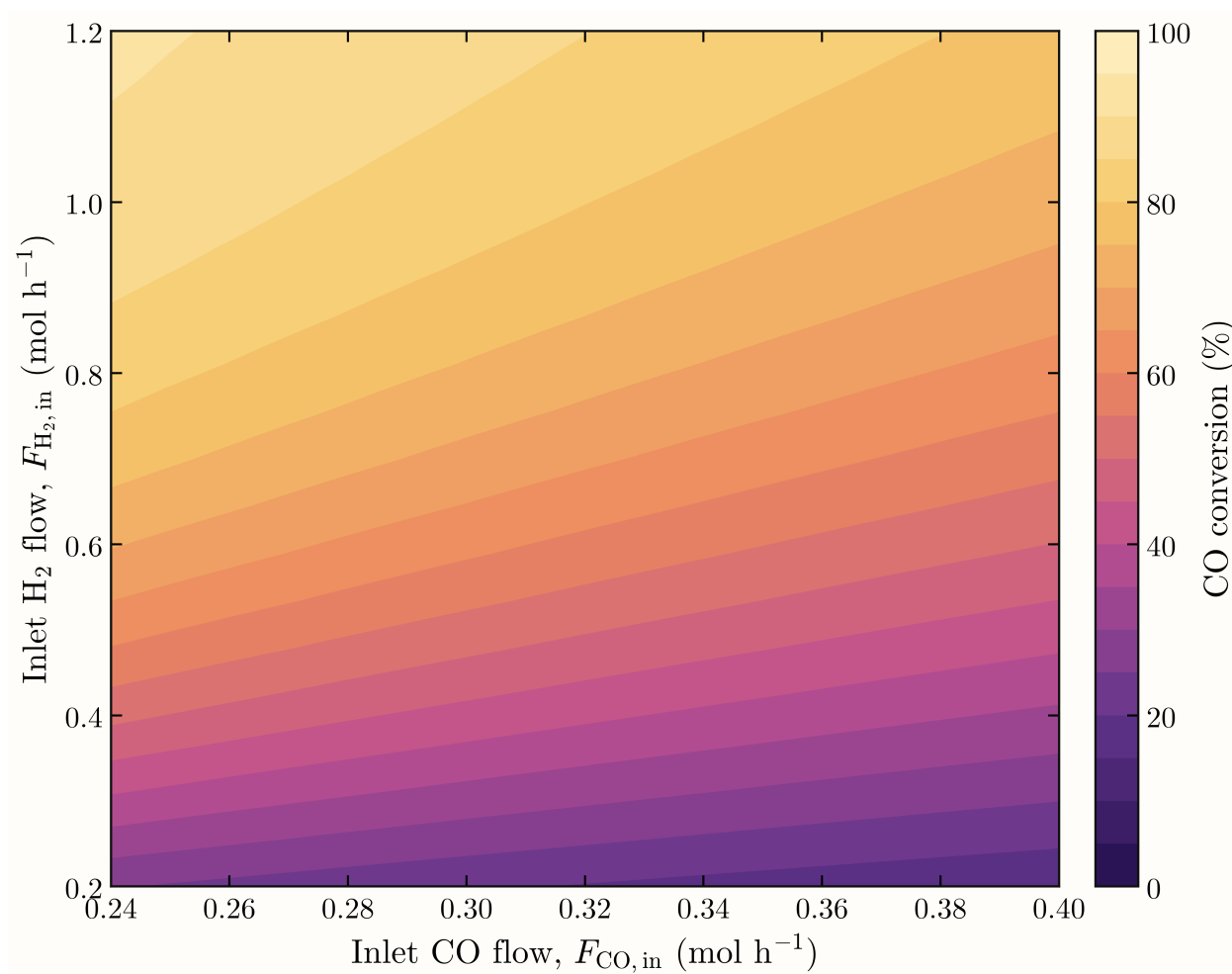


Figure 6. Full M4 predicted outlet CO conversion versus absolute inlet H₂ and CO flow; N₂ is fixed at 0.799 mol h⁻¹.

Each point is a no-fit Full M4 simulation at 350 °C, 1.00 bar, and 3.12 g catalyst under isothermal, constant-pressure conditions. Since the axes are absolute molar flows, moving across the map changes both feed composition and total throughput; this is not solely an H₂/CO-ratio sweep. The 1-bar condition is below the 5–15-bar kinetic-fit pressure range, so the map is an extrapolative model sensitivity rather than experimental validation. No experimental measurements are plotted.

CO conversion under nitrogen dilution

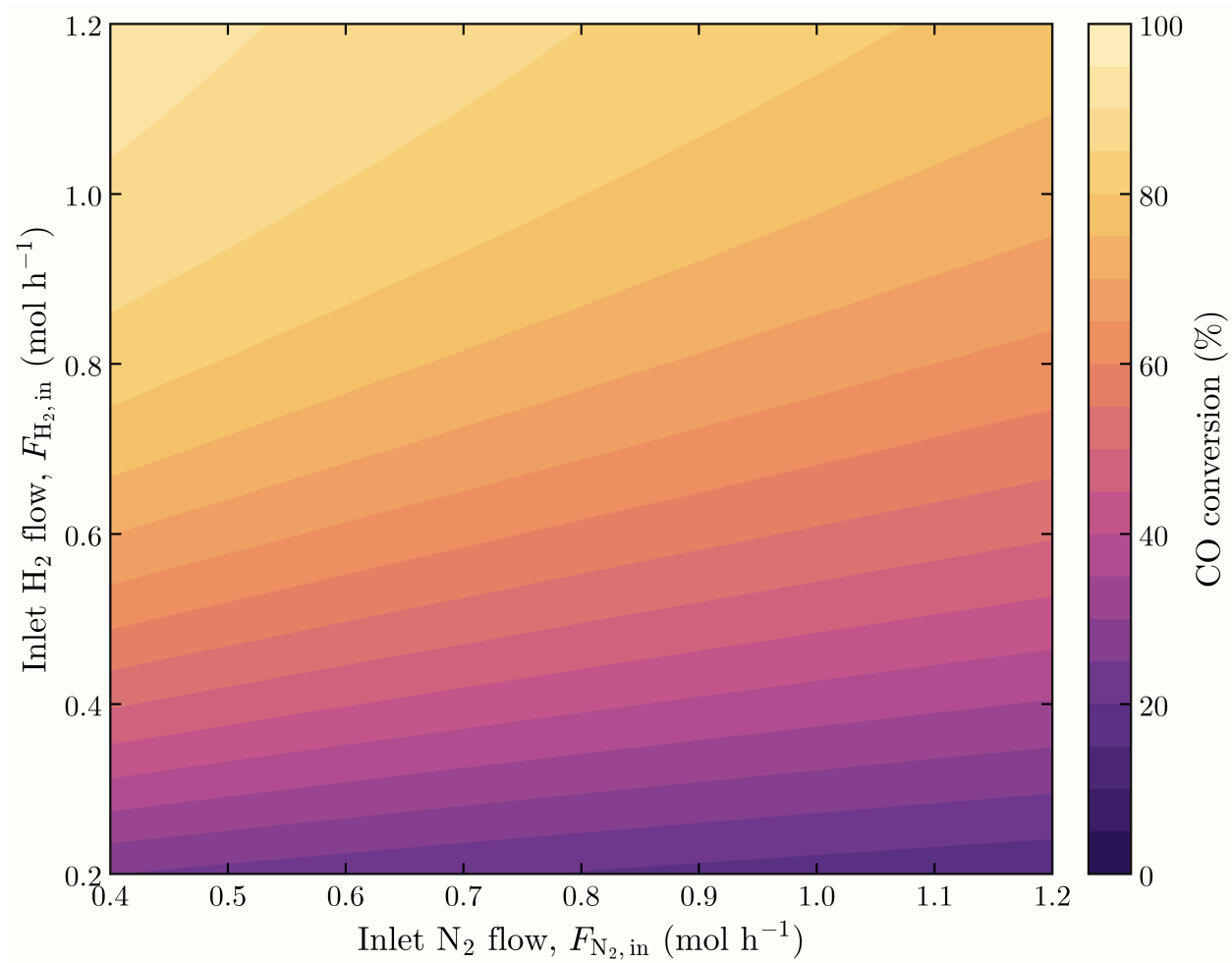


Figure 7. Full M4 predicted outlet CO conversion versus inlet H₂ and N₂ flow; CO is fixed at 0.320 mol h⁻¹.

This sweep isolates the conditional effect of nitrogen dilution while varying H₂ feed, with CO held fixed. At a given H₂ flow, increasing the inert N₂ feed lowers the reactant mole fractions and increases total throughput, reducing the predicted conversion. Conditions are 350 °C, 1.00 bar, and 3.12 g catalyst, isothermal and constant-pressure. As in Fig. 6, the 1-bar result extrapolates below the 5–15-bar kinetic-fit range and is not experimental validation; no experimental measurements are plotted.

Optimization

The discrete grid search identifies the best sampled operating point for the Full M4 reactor model. It maximizes completed-bed CO conversion over four inputs: inlet H₂/CO ratio, inlet temperature, inlet pressure, and catalyst space time on a CO-feed basis. The Experiment 1 catalyst mass (3.12 g), bed geometry and voidage, and N₂/CO inlet ratio are fixed. At each selected H₂/CO ratio and space time, the inlet flows are reconstructed as

$$F_{\text{CO},\text{in}} = \frac{W_{\text{cat}}}{\tau_{\text{cat},\text{CO}}}, \quad F_{\text{H}_2,\text{in}} = r_{\text{H}_2/\text{CO}} F_{\text{CO},\text{in}}, \quad F_{\text{N}_2,\text{in}} = \left(\frac{F_{\text{N}_2}}{F_{\text{CO}}} \right)_{\text{Exp1}} F_{\text{CO},\text{in}}.$$

Thus changing space time scales the inlet throughput while preserving N₂/CO; N₂ is not an independent optimization variable. Each point is an isothermal Full M4 simulation with Ergun pressure drop, solved using BDF. The grid contains 12 H₂/CO ratios, 9 inlet temperatures, 9 pressures, and 21 space-time values, for $12 \times 9 \times 9 \times 21 = 20,412$ simulations; all completed. The objective is

$$\max_{r_{\text{H}_2/\text{CO}}, T_{\text{in}}, P_{\text{in}}, \tau_{\text{cat},\text{CO}}} X_{\text{CO},\text{out}}, \quad X_{\text{CO},\text{out}} = 100 \left(1 - \frac{F_{\text{CO},\text{out}}}{F_{\text{CO},\text{in}}} \right) \%.$$

Table 1. Best sampled point in the four-factor grid search.

Quantity	Best sampled value	Position or role
Inlet H ₂ /CO ratio	5.00 mol/mol	Upper grid bound
Inlet temperature	370 °C	Interior grid value
Inlet pressure	15.0 bar abs	Upper grid bound
Catalyst space time, $W_{\text{cat}}/F_{\text{CO},\text{in}}$	140.4 kg _{cat} s mol _{CO} ⁻¹	Upper grid bound
CO / H ₂ / N ₂ inlet flows	0.080 / 0.400 / 0.19975 mol/h	Derived values
Outlet CO conversion	99.9999928%	Maximized objective
Outlet CH ₄ yield	99.9999318%	Predicted output

The maximum is the best sampled grid point, not a continuous or experimentally validated optimum. The selected point lies at the upper bounds for H₂/CO, pressure, and space time. The objective does not penalize hydrogen use, low throughput, or operating pressure, and the model is isothermal. Treat the result as exploratory; a practical operating recommendation would require those constraints and objectives to be defined.

Independent N₂-flow sensitivity

Figure 8 varies N₂ alone while holding the Experiment 1 CO and H₂ inlet molar flows fixed. This separate sweep is not part of the four-factor optimization, which keeps N₂/CO fixed. As N₂ rises, reactant dilution and total inlet flow both increase, so the curve represents their combined effect under otherwise fixed inputs.

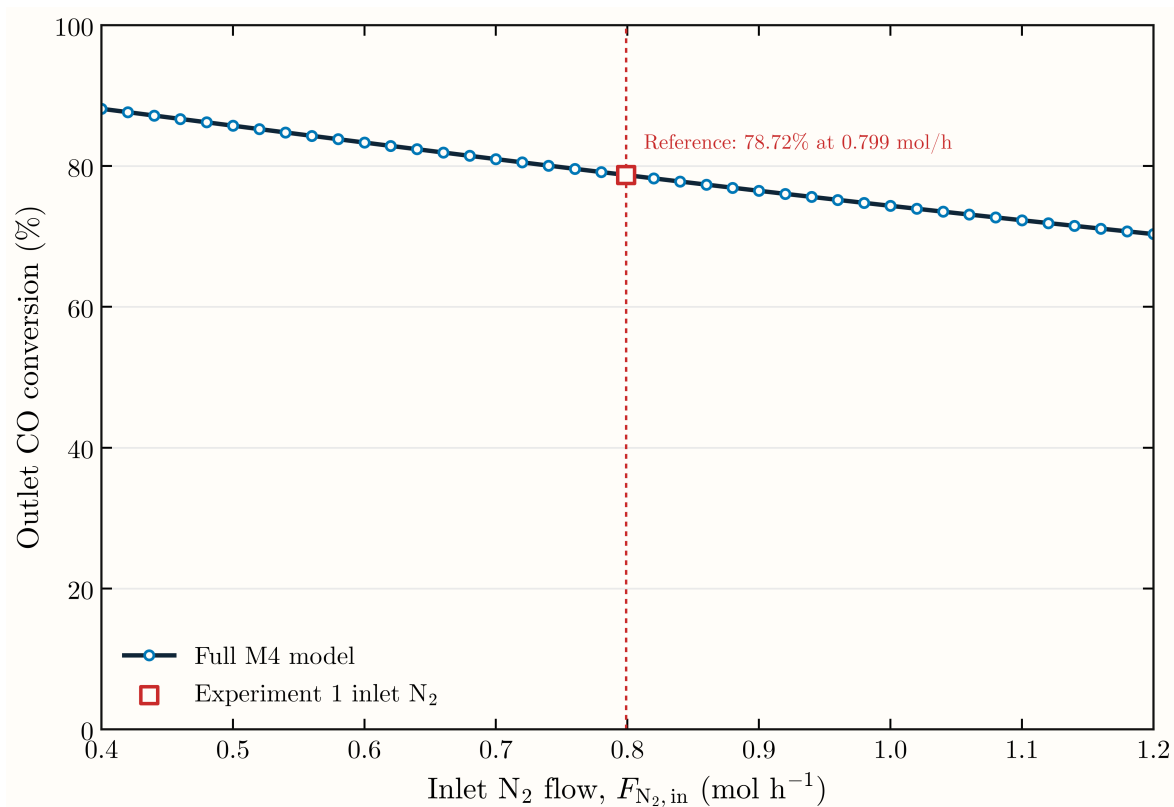


Figure 8. Full M4 predicted outlet CO conversion as inlet N₂ flow varies from 0.4 to 1.2 mol/h, with CO = 0.320 mol/h and H₂ = 0.959 mol/h fixed. The reference point at N₂ = 0.799 mol/h gives 78.72% conversion. Conditions are 350 °C, 1 bar absolute, and 3.12 g catalyst, isothermal and constant-pressure.

Conversion decreases from 88.14% to 70.34% over the plotted N₂ range. The 1-bar calculations are below the 5–15-bar kinetic-fit pressure range; interpret this as a conditional model sensitivity, not experimental validation.

Four-factor optimization map

The pressure-slice figure shows the conversion landscape around the selected grid point. Each panel varies inlet H_2/CO ratio and temperature at fixed pressure, with catalyst space time held at its optimized grid value. The star marks the best sampled point at 15 bar and 370 °C.

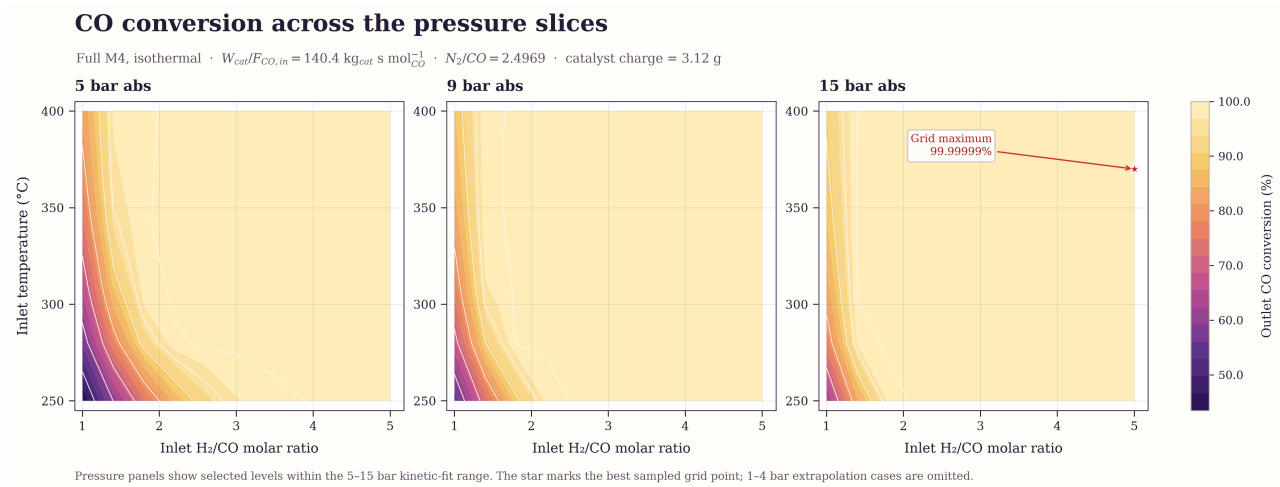


Figure 9. Pressure slices of predicted outlet CO conversion from the four-factor grid. Space time is fixed at $140.4 \text{ kg}_{\text{cat}} \text{ s mol}_{\text{CO}}^{-1}$; the star marks the best sampled point.

Only 5, 9, and 15 bar are shown because 1–4 bar fall below the kinetic-fit pressure range. The shared color scale allows comparison across panels. The figure and table report model predictions, not experimental validation.